# ACADEMIC PROJECT WORK REPORT

*On*

***"REINFORCEMENT LEARNING-BASED IDENTIFICATION OF FAILURE SCENARIOS FOR CONCEPT: DYNAMIC RISK ASSESSMENT OF AI-CONTROLLED ROBOTIC SYSTEMS***

***(3D Simulation for Training & Testing Robots)"***

*A report submitted in partial fulfillment of the requirements for the Award of Degree of*

**B.Tech in Department of Information Technology (IT)**

(Duration of Project Work: 01st January 2026 – 30th June 2026)

By

**Vishnu Vishwakarma (Roll No: 22354)**

**Institute of Engineering & Technology (IET),**

**Dr. Ram Manohar Lohia Avadh University (RMLAU),**

**Faizabad, Ayodhya – 224001 (U.P.)**

Under the supervision of

**Dr. Amit Kumar Sinha**

**(Supervisor & Assistant Professor)**

**School of Mechanical Engineering (ME), Faculty of Engineering**

**Shri Mata Vaishno Devi University (SMVDU),**

**Kakryal, Katra – 182320 (J&K)**

RESEARCH ARTICLE

3D Simulation for Training & Testing Robots

# REINFORCEMENT LEARNING-BASED IDENTIFICATION OF FAILURE SCENARIOS FOR CONCEPT: DYNAMIC RISK ASSESSMENT OF AI-CONTROLLED ROBOTIC SYSTEMS

## I.T. specialization in Data Science and Applications

**Vishnu Vishwakarma**

Department of Information Technology**,** Institute of Engineering & Technology (RMLAU) Faizabad, Ayodhya

**Dr. Amit Kumar Sinha**

School of Mechanical Engineering, Shri Mata Vaishno Devi University, Katra, Jammu & Kashmir

## Abstract

AI-controlled robotic systems can pose significant risks to both human workers and the environment due to their complex, autonomous operations. Traditional risk assessment methods fall short in adequately describe sophisticated failure scenarios of systems controlled by black-box policies. These traditional methods are applied manually and do not considering frequent software updates. Therefore, novel methods that enable a dynamic risk assessment for AI-controlled robotic systems are required. In this paper, we present the concept of a new approach that enables the dynamic risk assessment of unknown or dynamically changing control policies. This includes AI-controlled robotic systems, especially those utilizing reinforcement learning. The new approach combines two key methods: (i) automated identification of failure modes and (ii) a reinforcement learning-based search for critical failure scenarios. Additionally, it incorporates an evaluation method that analyzes the detected sets of critical scenarios or events. This evaluation method creates dynamic high-level risk models and identifies vulnerabilities to specific failures.

AI-controlled robotic systems pose a risk to human workers and the environment. Classical risk assessment methods cannot adequately describe such black box systems. Therefore, new methods for a dynamic risk assessment of such AI-controlled systems are required. In this paper, we introduce the concept of a new dynamic risk assessment approach for AI-controlled robotic systems. The approach pipelines five blocks: (i) a Data Logging that logs the data of the given simulation, (ii) a Skill Detection that automatically detects the executed skills with a deep learning technique, (iii) a Behavioral Analysis that creates the behavioral profile of the robotic systems, (iv) a Risk Model Generation that automatically transforms the behavioral profile and risk data containing the failure probabilities of robotic hardware components into advanced hybrid risk models, and (v) Risk Model Solvers for the numerical evaluation of the generated hybrid risk models.

Genetic Algorithms (G.A.s) are utilized to optimize fault injection parameters and automatically identify critical failure scenarios in AI-controlled robotic systems. This enables efficient exploration of fault combinations and thereby enhances/improving the effectiveness of dynamic risk assessment by discovering high-risk system behaviors.

## Keywords

Dynamic Risk Assessment, Hybrid Risk Models, Fault Injection, M2M (Model-to-Model) Transformation, ROS (Robot Operating System), AI-Controlled Robotic Systems, Genetic Algorithms, Deep Learning, Large Language Models (LLMs) & Reinforcement Learning

# 1. Introduction

Robotic systems can be classified as safety-critical due to their operation in close proximity to humans and frequent interactions with human operators. Such systems often incorporate moving components that could pose risks to human workers or handle hazardous substances, such as chemical mixtures. Robotic systems controlled by AI algorithms, especially reinforcement learning, have been a topic of interest for some time [1]. Recently, the application of reinforcement learning has experienced a significant increase in popularity [2]. Despite its growing adoption, traditional reinforcement learning lacks inherent safety guarantees. Dynamically changing control policies are essential for flexible production systems with high reconfigurability and repurpose-ability. They allow robotic systems to adapt to varying operational conditions and requirements, ensuring optimal performance. This adaptability enhances efficiency and enables rapid responses to changing production demands, making the manufacturing process more resilient and versatile. However, each new control policy can alter the behavior of the robotic systems, necessitating a renewed risk assessment. This underscores the need for a paradigm shift in risk assessment methodologies—from a static, one-time assessment prior to deployment to a dynamic risk assessment approach that continuously evaluates potential risks throughout the lifecycle of the system. The pivotal research question that emerges is: How can we dynamically assess the potential risks of robotic systems controlled by black-box policies?

Robotic systems can be safety-critical because they are often deployed in close proximity to human workers. Moreover, they collaborate with human workers. They may contain moving parts that can pose risks to human workers or carry substances dangerous to humans, such as chemical mixtures, as shown in Figure 1. Robotic systems are increasingly being controlled by AI algorithms such as Reinforcement Learning [1]. This requires a shift of the paradigm of risk assessment towards a dynamic risk assessment instead of the classical approach when the risk assessment of the system is done only once before the operation. The research question to be answered is: How to automatically and dynamically estimate the potential risk of robotic systems controlled by black-box policies?

To efficiently explore a large search space of potential fault conditions, Genetic Algorithms (GAs) can be incorporated as an optimization technique. Genetic Algorithms mimic the process of natural evolution to identify optimal combinations of fault parameters, such as fault type, fault location, fault duration, and fault magnitude. By evolving candidate fault scenarios over multiple generations, GAs can efficiently discover critical and high-risk failure conditions that may not be detected through conventional testing approaches. The integration of Genetic Algorithms with Reinforcement Learning and Fault Injection techniques can significantly enhance the identification of hazardous robotic behaviors, thereby supporting dynamic risk assessment and improving the safety, reliability, and robustness of AI-controlled robotic systems.

**Contribution:** This paper introduces the concept of a novel approach for dynamic risk assessment of control policies that are either unknown or subject to dynamic changes, with a particular focus on AI-controlled robotic systems that employ reinforcement learning. The proposed method is illustrated in Figure 1. This innovative approach integrates two principal techniques: automated identification of failure modes and a reinforcement learning-based search for critical failure scenarios. Additionally, it includes an evaluation method that analyzes the detected sets of critical scenarios or events. This method constructs dynamic high-level risk models and identifies vulnerabilities to specific failures. The applicability of the proposed approach will be evaluated in a real setup with a Franka Emika Panda robot and reinforcement learning policies.

This paper presents a concept of a new approach that enables the dynamic risk assessment of unknown or dynamic control policies including AI-controlled robotic systems. The proposed pipeline, shown in Figure 2, consists of five blocks:

- Data Logging consists of a hardware-interface integrated ROS logger. This block analyzes the in simulation executed ROS code. It logs time-series data such as the positions and velocities of each joint.

- Skill Detection block is a deep learning-based classifier. This classifier detects the executed skills and labels the time-series data with these skills e.g. move, pick, carry, etc.

- Behavioral Analysis block generates the behavioral profile of the given robotic systems. The behavioral profile contains information about the sequence of executed skills, active components for each skill, and properties of components such as velocity, execution time, intensity, etc. that influence reliability characteristics of the components.

- Risk Model Generation block transforms the behavioral profile and risk data, e.g. failure probabilities of robotic hardware components, into advanced hybrid risk models.

- Finally, the Risk Model Solver quantifies the generated risk models.

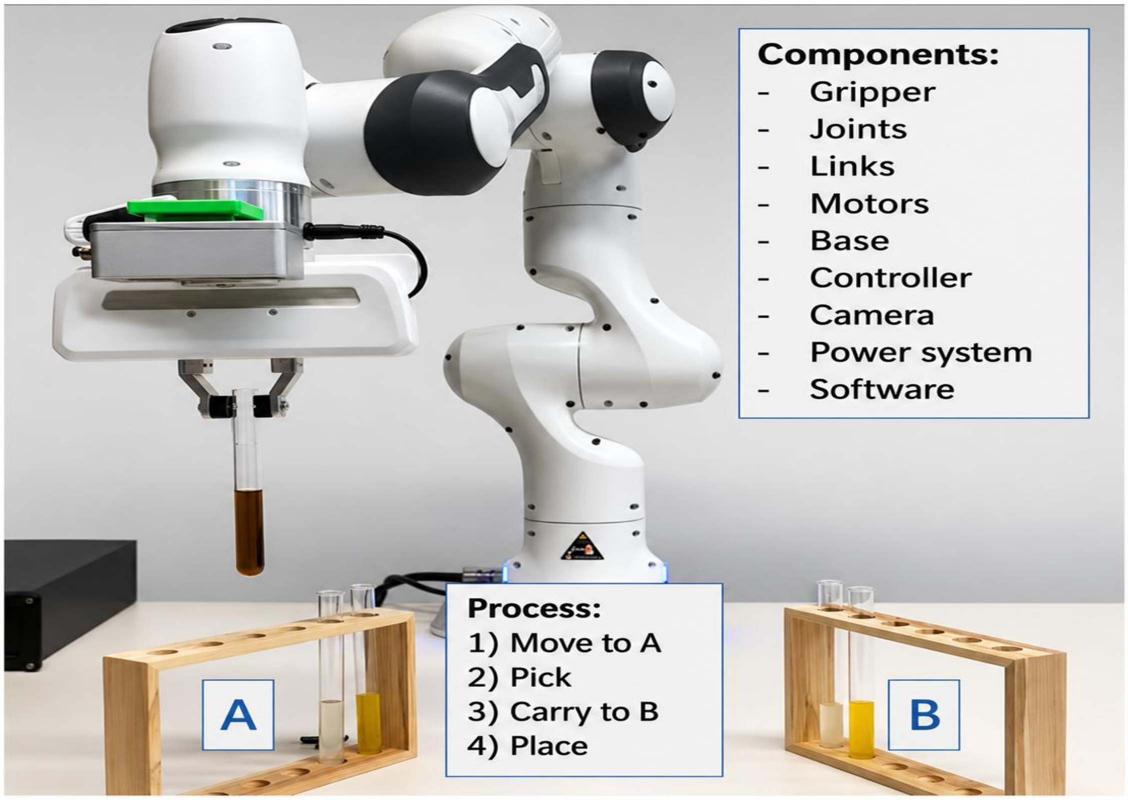



Figure 1. Robotic manipulator which has been trained to grasp a test tube at position A and place it to position B.

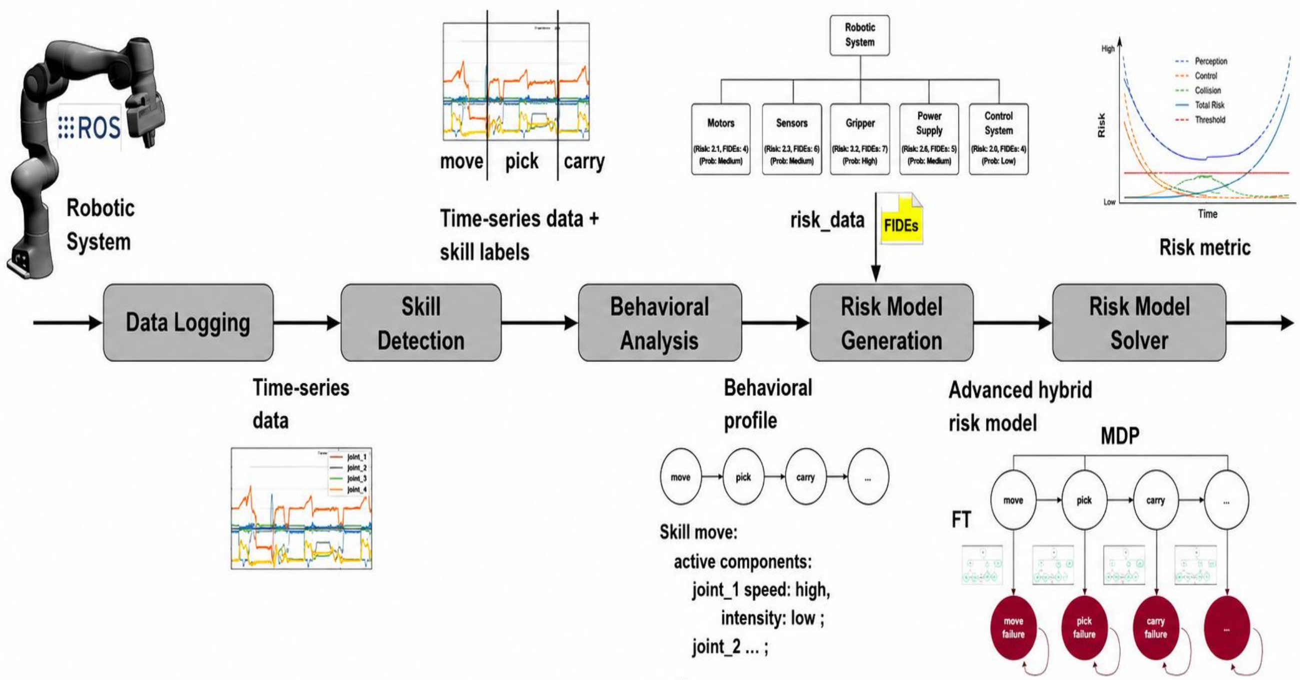



Figure 2. The proposed approach to the dynamic risk assessment of AI-controlled robotic systems.

## 2. STATE OF THE ART/Literature Review (Literature Survey)

This section discusses three relevant groups of methods, which are crucial for explaining the novelty of the proposed concept. First, dynamic reliability and risk assessment methods for robotic systems. Second, large language models for risk assessment. Third, fault injection methods.

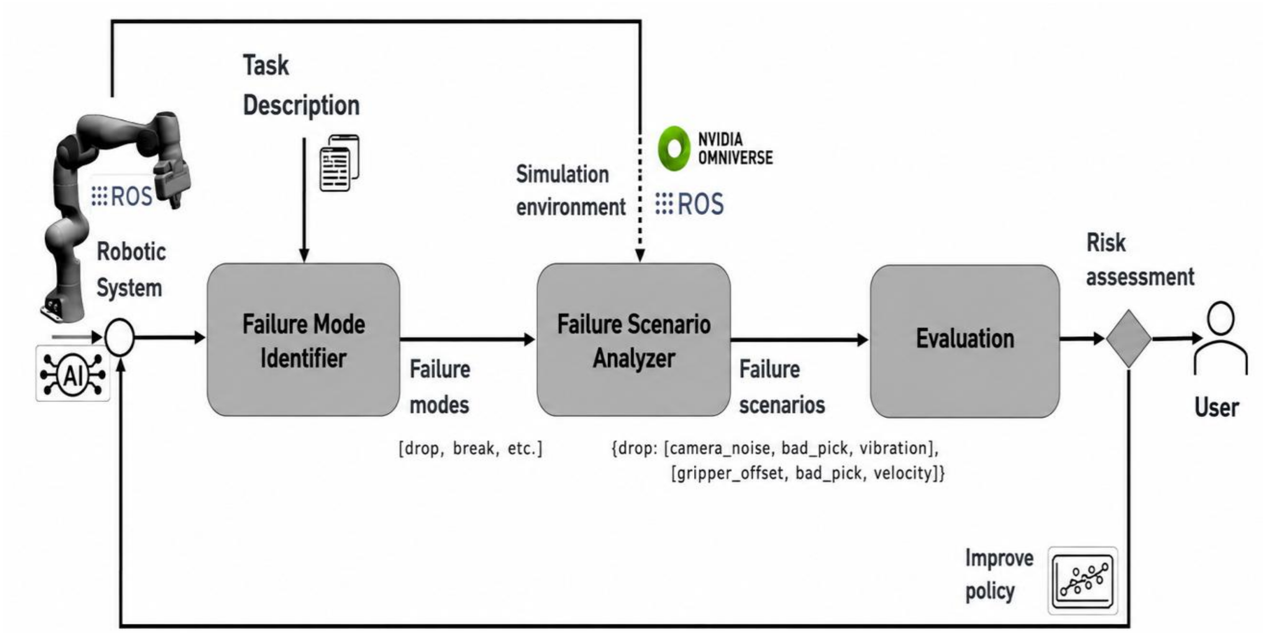



**FIGURE 1: The proposed approach to the reinforcement learning-based identification of failure scenarios for dynamic risk assessment of AI-controlled robotic systems.**

### *2.1 Dynamic Risk Assessment*

Dynamic reliability and risk assessment methods are essential for understanding complex failure scenarios in modern robotic systems. A diverse range of researchers has explored dynamic risk assessment, employing various risk models and applying them to specific contexts. For instance, in [3] and [4], Bayesian networks and Bayesian theory are utilized to dynamically update failure probabilities, as shown in case studies with storage tanks containing hazardous chemicals and water control systems. Schneider [5] provides an overview of dynamic risk management for cyber-physical systems, while Kaul et al. [6] focus on the reliability of intelligent mechatronic systems, transforming system models into Bayesian networks, especially beneficial during early design stages. The Situation-Aware Dynamic Risk Assessment (SINADRA) framework [7] specifically targets autonomous systems, using Bayesian networks for modeling and integrating situation-specific risk features.

In the context of Human-Robot Collaboration (HRC), Vicentini et al. [8] and Inam et al. [9] discuss safety and risk, with the former automating traditional risk analysis methods through formal model verification, and the latter integrating Hazard Operability (HAZOP) techniques with risk estimation. Our previous work [10–12] introduced automated and continuous risk assessment methods for software-defined systems, including robotics and manufacturing, utilizing transformation algorithms from SysML v2 and ROS code to hybrid risk models.

Dong et al. [13] presented a dependability analysis for deep reinforcement learning in robotics and autonomous systems using Discrete-Time Markov Chain (DTMC) modeling and Probabilistic Model Checking (PMC) to assess safety, robustness, and resilience. Brunke [14] reviews safe learning control approaches, noting the limitations of standard reinforcement learning in safety assurance, while highlighting the potential of safe model-based reinforcement learning despite its current practical constraints. Kwon et al. [15] explored how learning models generated through reinforcement learning can balance rewards with the fulfillment of safety constraints, illustrated in a case study involving delivery robots. The approach introduced by Camilli et al. (2021) [16] presents a risk modeling method specifically designed for collaborative AI systems operating alongside humans in shared spaces to achieve common goals. This risk model incorporates goals, risk events, and domain-specific indicators that may expose humans to hazards, enhancing assurance methods by leveraging insights from run-time evidence. The method's effectiveness is demonstrated through an Industry 4.0 example, where a machine learning-equipped robotic arm collaborates with a human operator. Villani et al. (2018) [17] provide an in-depth examination of safety standards and intuitive interfaces tailored for collaborative robotics in industrial settings, emphasizing safety issues, user interfaces, and practical applications.

Our review of dynamic reliability and risk assessment literature, particularly focusing on reinforcement learning-controlled systems, indicates a gap in dynamic risk assessment for systems controlled by black-box policies and the automated identification of critical failure scenarios. Our approach aims to fill these gaps by enabling the evaluation of vulnerabilities and overall risk assessment for AI-controlled robotic systems, addressing the identified limitations in the current state of the art.

### *2.2 Large Language Models for Risk Assessment*

Large Language Models (LLMs) [18], such as the Generative Pre-trained Transformer (GPT) [19] and Bidirectional Encoder Representations from Transformers (BERT) [20], have demonstrated exceptional performance across a variety of Natural Language Processing (NLP) tasks. Research by Qi et al. [21] investigated the use of ChatGPT for applying Systems Theoretic Process Analysis (STPA) to Automatic Emergency Brake (AEB) and Electricity Demand Side Management (DSM) systems. Their findings suggest that relying solely on ChatGPT without human expert intervention leads to challenges due to the unreliable nature of LLMs. However, a combined approach involving both humans and ChatGPT could potentially surpass the capabilities of human experts alone in conducting STPA. Additionally, Diemert et al. [22] explored the integration of LLMs in hazard analysis for safety-critical systems through a process they termed co-hazard analysis (CoHA). In CoHA, a human expert engages with an LLM in a context-aware chat session, leveraging the interaction to identify potential hazards. Their experimental results indicate that while CoHA can be effective for STPA in simpler systems, its efficacy diminishes with increasing system complexity.

### *2.3 Fault Injection*

Significant advancements have been made in fault injection techniques using ROS and a simulation platform known as Gazebo for safety-critical robotic applications. Specifically, Favier et al. [23] developed a ROS/Gazebo-based hierarchical fault-tolerant framework for autonomous mobile robots that enables fault injection and the analysis of error propagation chains. Zhao et al. [24] explored sensor faults in mobile robot localization systems, albeit with relatively simplistic fault parameters. Morais et al. (2015) [25] introduced a fault diagnosis method for multiple mobile robots, implementing the fault injector module via the ROS service.

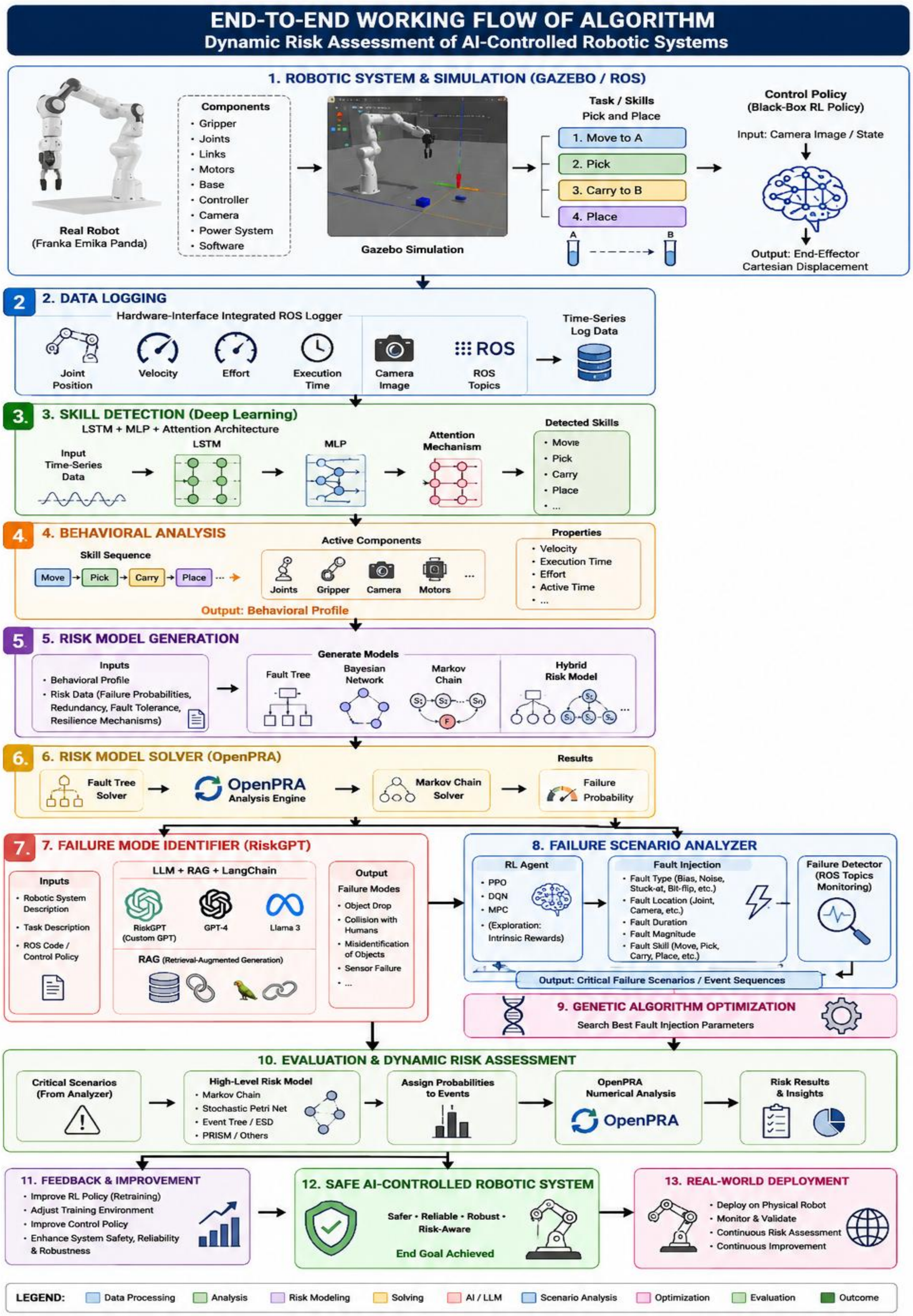
Hsiao et al. [26] presented MAVFI, an end-to-end fault analysis framework that offers an extensive evaluation of fault injection across various algorithms, detailing error propagation and recovery strategies within a simulation setting. MAVFI utilizes a ROS node that leverages the ROS communication protocols and Linux system commands to facilitate fault injection. In [27], a model-based fault injection method, FIBlock, was extended with Deep Reinforcement Learning capabilities. This method, implemented in Simulink, enables the automated search for fault parameters that result in the most significant system response.

Additionally, there are specialized ROS-based fault injection methods and tools designed for robotic manipulators. Alemzadeh et al. [28] and Li et al. [29] have injected faults into teleoperated surgical robots within a simulation environment, focusing on the effects of malicious control commands from different types of cyberattacks and using a model-based framework to assess the consequences. Lastly, Ma et al. [30] proposed a fault injectionbased risk analysis method tailored to ROS, demonstrating its effectiveness through a case study involving a robot manipulator. This approach aids in pinpointing critical failures in robotic manipulators. We will adapt and extend the fault injecton methods presented in [30] and [27] to meet our specific requirements.

Several studies have demonstrated the effectiveness of Genetic Algorithms (GAs) in optimization and fault analysis problems. Genetic Algorithms are widely used to explore large search spaces and identify optimal parameter combinations. In robotic systems, GAs have been applied to optimize fault injection parameters and discover critical failure scenarios. Their ability to efficiently search complex fault combinations makes them a promising technique for enhancing dynamic risk assessment and improving the safety and reliability of AI-controlled robotic systems.

## 3. CONCEPT/Proposed Methodology

The proposed concept is depicted in Figure 1 and discussed in the following three subsections. The robotic manipulator as shown in Figure 1 has been trained to grasp a test tube at position A and place it to position B. The robotic manipulator uses the skills Move to A, Pick, Carry to B, and Place. The real robotic manipulator consists of several components such as Gripper, Joints, Links, Motors, Base, Controller, Camera, Power system, and Software.



END-TO-END WORKING FLOW OF ALGORITHM
Dynamic Risk Assessment of AI-Controlled Robotic Systems
1. ROBOTIC SYSTEM & SIMULATION (GAZEBO / ROS)
Real Robot (Franka Emika Panda)
Components
• Gripper
• Joints
• Links
• Motors
• Base
• Controller
• Camera
• Power System
• Software
Gazebo Simulation
Task / Skills
Pick and Place
1. Move to A
2. Pick
3. Carry to B
4. Place
A
B
Control Policy (Black-Box RL Policy)
Input: Camera Image / State
Output: End-Effector Cartesian Displacement
2. DATA LOGGING
Hardware-Interface Integrated ROS Logger
Joint Position
Velocity
Effort
Execution Time
Camera Image
ROS
ROS Topics
Time-Series Log Data
3. SKILL DETECTION (Deep Learning)
LSTM + MLP + Attention Architecture
Input Time-Series Data
LSTM
MLP
Attention Mechanism
Detected Skills
• Movie
• Pick
• Carry
• Place
• ...
4. BEHAVIORAL ANALYSIS
Skill Sequence
Move
Pick
Carry
Place
Active Components
Joints
Gripper
Camera
Motors
Properties
• Velocity
• Execution Time
• Effort
• Active Time
• ...
Output: Behavioral Profile
5. RISK MODEL GENERATION
Inputs
• Behavioral Profile
• Risk Data (Failure Probabilities, Redundancy, Fault Tolerance, Resilience Mechanisms)
Generate Models
Fault Tree
Bayesian Network
Markov Chain
Hybrid Risk Model
6. RISK MODEL SOLVER (OpenPRA)
Fault Tree Solver
OpenPRA Analysis Engine
Markov Chain Solver
Results
Failure Probability
7. FAILURE MODE IDENTIFIER (RiskGPT)
Inputs
• Robotic System Description
• Task Description
• ROS Code / Control Policy
LLM + RAG + LangChain
RiskGPT (Custom GPT)
GPT-4
Llama 3
RAG (Retrieval-Augmented Generation)
Output
Failure Modes
• Object Drop
• Collision with Humans
• Misidentification of Objects
• Sensor Failure
• ...
8. FAILURE SCENARIO ANALYZER
RL Agent
• PPO
• DQN
• MPC
• (Exploration: Intrinsic Rewards)
Fault Injection
• Fault Type (Bias, Noise, Stuck-at, Bit-flip, etc.)
• Fault Location (Joint, Camera, etc.)
• Fault Duration
• Fault Magnitude
• Fault Skill (Move, Pick, Carry, Place, etc.)
Failure Detector (ROS Topics Monitoring)
Output: Critical Failure Scenarios / Event Sequences
9. GENETIC ALGORITHM OPTIMIZATION
Search Best Fault Injection Parameters
10. EVALUATION & DYNAMIC RISK ASSESSMENT
Critical Scenarios (From Analyzer)
High-Level Risk Model
• Markov Chain
• Stochastic Petri Net
• Event Tree / ESD
• PRISM / Others
Assign Probabilities to Events
OpenPRA Numerical Analysis
OpenPRA
Risk Results & Insights
11. FEEDBACK & IMPROVEMENT
• Improve RL Policy (Retraining)
• Adjust Training Environment
• Improve Control Policy
• Enhance System Safety, Reliability & Robustness
12. SAFE AI-CONTROLLED ROBOTIC SYSTEM
Safer • Reliable • Robust • Risk-Aware
End Goal Achieved
13. REAL-WORLD DEPLOYMENT
• Deploy on Physical Robot
• Monitor & Validate
• Continuous Risk Assessment
• Continuous Improvement
LEGEND:
Data Processing
Analysis
Risk Modeling
Solving
AI / LLM
Scenario Analysis
Optimization
Evaluation
Outcome

### 3.1. Data Logging

The input of the Data Logging is a Gazebo simulation that executes a control policy, in the form of any ROS code that utilizes the hardware abstractions. The control policy could be the policy that controls the robotic manipulator to grasp a test tube at position A and place it to position B, as shown in Figure 1.

The hardware-interface integrated ROS logger publishes messages about active robotic components. These ROS topics in Gazebo, include among others, information about the position, velocity, effort, and execution time of robotic components. More specifically, we log the positions and speeds of each joint. The log contains all this information as time-series data. The logger can log data from any ROS code independent of the programming language (Python or C++) and ROS version (ROS1 or ROS2). The logger is not hardware-specific and can log the data of any ROS-controlled robotic policy. To map the hardware components directly to risk data, the logger demands hardware descriptions to be published inside the existing hardware abstractions.

### 3.2. Skill Detection

The deep learning-based skill detector developed for robotic manipulators is one of the key elements of this approach. It labels the logged time-series data with executed skills. Executed skills could be move, pick, carry, etc. The model shall be trained on meticulously self-generated data sets, including training and test data. Therefore, the skill detection can reliably detect and classify robot skills. We generate the data set from various robotic control algorithms. We arbitrarily choose the parameters, such as speed, execution times, path planning, etc. That gives us a particularly comprehensive data set. Preliminary the detector should combine Long Short-Term Memory (LSTM), Multilayer Perceptron (MLP), and possibly attention based architectures.

### 3.3. Behavioral Analysis

The Behavioral Analysis block interprets the time-series data with skill labels. To interpret the time-series data, we will use a rule-based approach. The Behavioral Analysis block creates the sequence of executed skills with exact time periods. In addition, it maps active components to the corresponding skills and adds available properties to the component. The Behavioral Analysis creates and outputs the behavioral profile of the system. It contains a list of skills (e.g. move, pick, carry, place, etc.) with exact execution time, a list of active components (joints, torque sensors, camera, gripper, etc.) during each skill, and the corresponding properties e.g., velocity, active time, effort, etc. for each component. The properties of each component can be different for each skill.

### 3.4. Risk Model Generation

The Risk Model Generation block parses the behavioral profile and risk data. The risk data contains failure probabilities, redundancy information, and other fault tolerance and resilience mechanisms of robotic hardware components. This information has to be added by experts. The transformation algorithm creates a fault tree [2] for each detected skill. Basic events of the fault tree are the failures of system components that are active during the execution of the skill. The logic of the fault tree represents potential fault tolerance mechanisms such as redundancies, recoveries, spares, etc. In case of no fault tolerance the top event is the failure of the skill modeled as an OR-gate. If the algorithm detects a redundant component, it creates an AND-gate and adds the corresponding quantity of the components as basic events. The failure probabilities of the basic events can vary depending on the logged execution time, velocity, intensity, etc. Reliability block diagrams [3] could be used as an alternative to fault trees. In case of dependent failures, fault tree models can be combined with Bayesian networks [4].

According to the sequence of executed skills, the transformation algorithm creates a dynamic high level risk model based on a Markov chain [5], stochastic Petri net [6], dual graph error propagation model [7], or PRISM language [8]. In case of a simple discrete-time Markov chain model, it creates a transient state for each skill and a corresponding absorbing failure state. An absorbing ”done” state is created, which serves as a state of success. The transition probability from a state to its corresponding failure state is defined by interconnected fault trees. The output of the transformation algorithm is a hybrid risk model that contains a Markov chain with interconnected fault trees in the OpenPRA model exchange format.

### 3.5. Risk Model Solver

We use the OpenPRA framework [9] for numerical risk assessment. OpenPRA is an open-source framework, which integrates multiple risk methods into an easy-to-use environment. The OpenPRA integrated analysis module can analyze hybrid risk models, such as Markov chains with interconnected fault trees. It first solves the fault trees by calling the fault tree analysis solver. The computed results are added to the transitions of the Markov chain. The Markov chain solver solves afterward the final Markov chain.

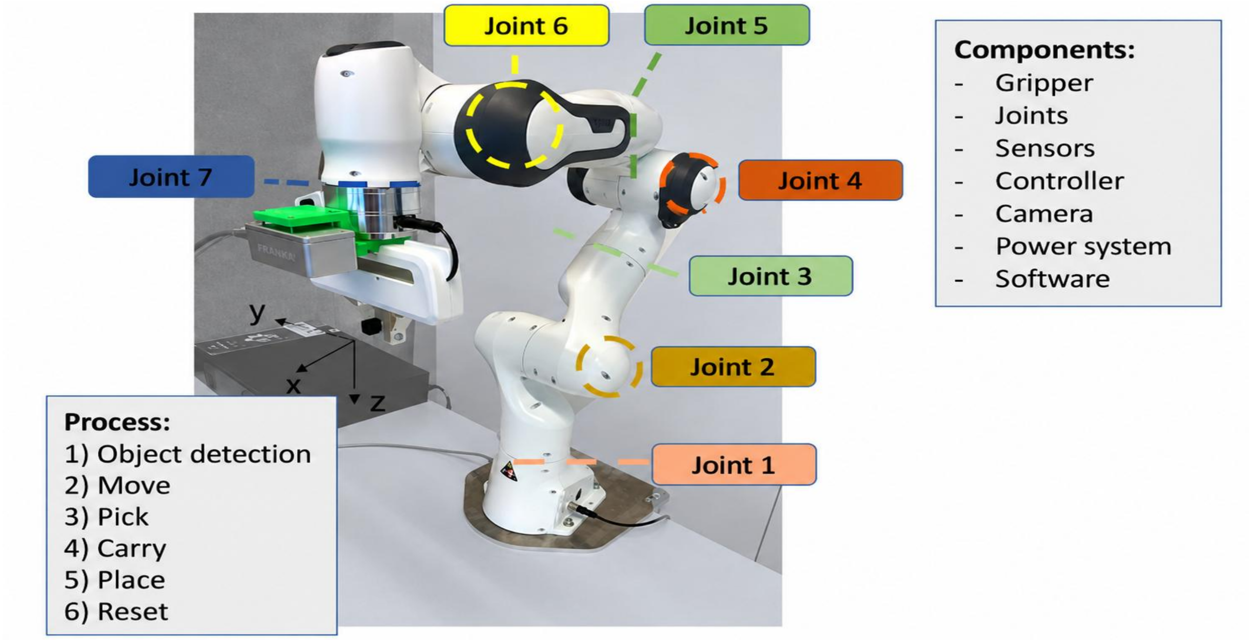



**FIGURE 4: Robotic manipulator which has been trained to detect, grasp, and place an object.**

### *3.6. Failure Mode Identifier*

The Failure Mode Identifier, illustrated in Figure 2, will employ machine learning techniques, including large language models and decision trees, to automate the identification of failure modes in robotic systems. Our investigation will be extended to incorporate Meta's latest Llama 3 model and OpenAI's GPT4 model, examining their capabilities in enhancing our current methodologies for failure mode identification. This will allow us to evaluate the effectiveness of state-of-the-art language models in identifying failure modes in robotic systems.

In addition, we will also investigate Retrieval-Augmented Generation (RAG) [31, 32] and LangChain [33] frameworks to further refine our approach. RAG will help us in integrating external knowledge sources during the failure mode identification process, thereby improving the accuracy and relevance of the identified failure modes. LangChain will facilitate the chaining of language model prompts, allowing for more sophisticated and context-aware interactions with the system.

The failure mode identifier will process inputs such as ROS code, which controls the robots actions, alongside detailed descriptions of the robotic system and its tasks. These descriptions will encompass the structural components of the robotic system, including joints, cameras, grippers, controllers, and power supply, providing a comprehensive understanding of the systems architecture. The task specifications will elaborate on the control policy, exemplified by a robotic system trained to detect, grasp, and place objects. The anticipated output from this method is an exhaustive list of potential failure modes that are specific to the robots designated task and system configuration. This list will be presented to the user, who will have the opportunity to confirm the accuracy of the findings and to supplement any additional failure modes as needed.

### *3.7. Failure Scenario Analyzer*

The Failure Scenario Analyzer, illustrated in Figure 3, will dynamically explore event sequences that could lead to critical failure modes. It utilizes a simulation environment, such as NVIDIA Isaac Sim [34], Gazebo [35], or MuJoCo [36] to execute control policies written in ROS code. The analyzer incorporates three main components: a Reinforcement Learning agent, a Fault Injection method, and a Failure Detector.

The Reinforcement Learning agent is the key element to this setup, determining actions based on observations and rewards. We will experiment with both model-free and model-based [37] reinforcement learning strategies. In model-free methods, we will use Proximal Policy Optimization (PPO) [38] to experiment with a stable on-policy algorithm and Deep Q-Learning [39] based algorithms to investigate the exploration advantages of off-policy algorithms. These algorithms will be accompanied by intrinsic exploration strategies such as random network distillation [40], so that the agent can explore and exploit potential critical failure scenarios within a feasible time frame. For model-based approaches, we plan to implement Model Predictive Control (MPC) [41], possibly enhanced by Monte Carlo Tree Search (MCTS) or Bayesian Optimization. These policies will be trained within simulation environments like Gazebo, NVIDIA Isaac Sim, or MuJoCo.

The Fault Injection method, adapted from [30] and [27], enables the injection of faults based on the Reinforcement Learning agents specified actions. These actions define the fault characteristics: fault type (such as bias, noise, stuck-at, packet loss, bit-flips), fault location (e.g., the camera image or a joint position signal), fault duration, fault skill during which the fault is injected (e.g., move, pick, carry, place, rotate), and fault magnitude, which indicates the intensity of the fault. Based on these configurations, faults are systematically introduced into the robotic system.

The Failure Detector monitors the systems response by subscribing to various ROS topics, thereby identifying the occurrence of any failure mode. This monitoring informs the agent’s decision-making process. Ultimately, the output from the Failure Scenario Analyzer is a set of critical scenarios or events that could result in severe failure modes, aiding in preemptive safety measures.

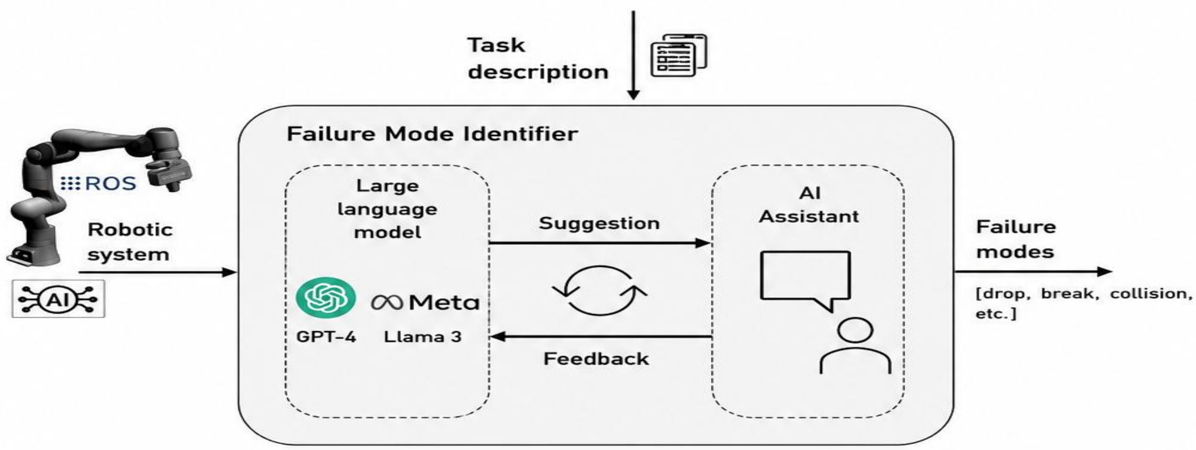



**FIGURE 2: The Failure Mode Identifier automatically identifies failure modes in robotic systems.**

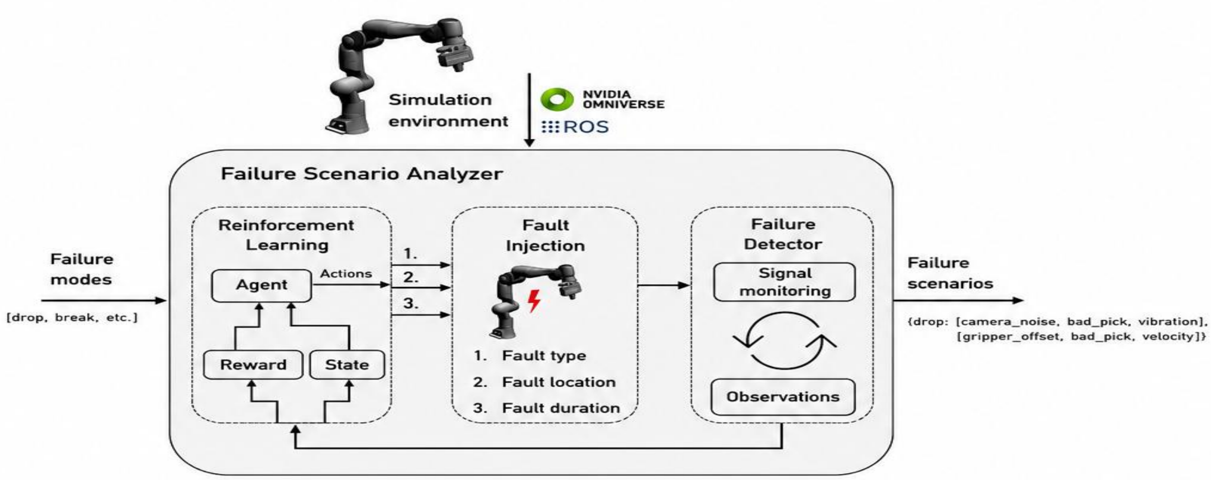



**FIGURE 3: The Failure Scenario Analyzer dynamically explores event sequences that could lead to critical failure modes.**

### *3.7.1. Role of Genetic Algorithms*

In addition to Reinforcement Learning, Genetic Algorithms (GAs) can be employed to optimize fault injection parameters and automatically discover critical failure scenarios in AI-controlled robotic systems. The GA searches a large space of fault combinations, including fault type, fault location, fault duration, and fault magnitude. By evaluating the severity of the resulting failures, the algorithm identifies the most critical scenarios that may lead to object drops, collisions, sensor malfunctions, or unsafe robotic behavior. This optimization process enhances dynamic risk assessment and supports the development of safer and more reliable reinforcement learning-based robotic systems.

### *3.8. Evaluation*

The evaluation method analyzes detected sets of critical scenarios or events. Based on the sequence of these events, the algorithm creates a dynamic high-level risk model. This model may employ various risk and reliability models, such as a Markov chains, stochastic Petri nets, event trees, event sequence diagrams, dual graph error propagation models, or PRISM. By assigning probabilities to the events within these models, we can quantitatively assess the likelihood of occurrence of different failure modes. We utilize the OpenPRA framework [42] for numerical risk assessment.

The evaluation method not only identifies vulnerabilities to specific failures but also provides actionable insights. These insights can be used to refine the training of the original reinforcement learning agent by adjusting the training environment so that critical failure scenarios are more likely to occur, thereby incentivizing their prevention. Additionally, the findings can inform software developers on how to enhance the control policy, thereby improving system robustness, reliability, and safety.

## 4. REAL-WORLD EXAMPLE/Experimental Setup

We will validate and show the applicability of our proposed approach in a real-world setup using a Franka Emika Panda robot, shown in Figure 4 and reinforcement learning policies.

### *4.1 Robotic System*

The robotic manipulator, as depicted in Figure 4, comprises several key components: Gripper, Joints, Sensors, Controller, Camera, Power System, and Software. We will conduct experiments in both virtual and real settings. In the virtual environment, platforms like Gazebo or NVIDIA Isaac Sim serve as the digital twin of the system, which we use to evaluate our methodology. This will later be validated in a real-world environment. To ensure consistency, the physical parameters of the manipulator in the virtual environment are set to mirror those of the actual robot.

### *4.2 Control Policy*

The robotic manipulator underwent training to perform the tasks of detecting, grasping, and placing objects. However, the specific control policy governing these actions remains undisclosed — a black-box policy, which in this context refers to a reinforcement learning algorithm. The expected process involves a sequence of skills including Object Detection, Move, Pick, Carry, and Place as depicted in Figure 4. This control policy, implemented through the reinforcement learning algorithm, will be executed within a simulation environment and will be analyzed with our proposed approach. During the training process of the reinforcement learning control policy, we experimented with both PPO and Deep Q-Learning based algorithms to fine-tune the black-box policy. While PPO operates on-policy, Deep Q-Learning based algorithms can work off-policy. Our tests revealed that Deep Q-Learning based algorithms exhibited promising potential for faster training; however, its stability was comparatively lower. In contrast, PPO demonstrated remarkable stability and consistently converged to the optimal policy for each training environment. This reliability made PPO the more practical choice for our purposes. The input to our agent is the raw camera image, and the output is the Cartesian displacement for the end-effector pose. We employed Sim2Real [43] process to transfer our trained policies from the Gazebo simulation environment to the physical Franka Emika Panda robot.

## 5. CHALLENGES AND PRELIMINARY RESULTS

In our research on reinforcement learning-based identification of failure scenarios for dynamic risk assessment, we encountered several challenges and achieved notable preliminary results, which are outlined below:

### *5.1 Challenges*

During our research on the Failure Scenario Analyzer, we identified several challenges that we are actively working to overcome. One challenge is ensuring simulation fidelity. When the failure detector is deployed in simulation, it can only detect scenarios within the scope of the simulation's fidelity. Given the constraints of available computational resources, a compromise between speed and accuracy is often necessary. Consequently, potentially identified failure scenarios can form as an efficient baseline for further validation through the physical setup to ensure their accuracy. Additionally, tuning and adapting Reinforcement Learning algorithms remains a challenge. It is crucial to enable these algorithms to perform continuous and effective exploration, including identifying multiple different failure causes. Achieving this requires careful calibration to balance exploration and exploitation, ensuring comprehensive identification of potential failure modes. These challenges highlight the complexity of developing a robust failure mode identifier and underscore the need for further research and innovation.

**Proposed Enhancement/Future Work/Future Scope:**

Future work will investigate the integration of Genetic Algorithms with Reinforcement Learning to improve the exploration of fault injection parameters and accelerate the discovery of critical failure scenarios.

### *5.2 Preliminary Results*

**Failure Mode Identifier:** We crated our own GPT model called RiskGPT. The goal of RiskGPT is to provide failure modes and severity levels based on the robotic system and task description. During the configuration, we uploaded the description of our Franka Emika Panda robot. After adding the detailed description of the control policy of our real world example, we obtained some initial results. The three most critical failure modes identified by our model are:

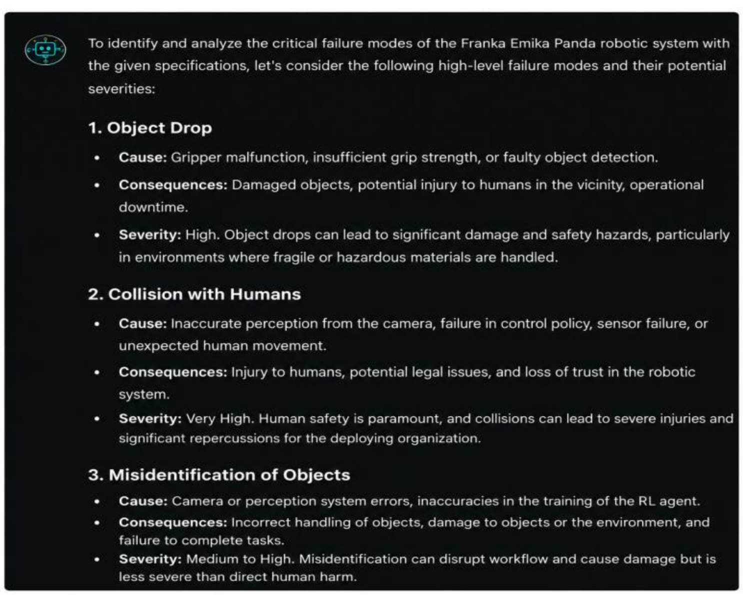
1. Object drop. 2. Collision with humans. 3. Misidentification of objects.

Figure 5 shows the prompt of our model. In addition, it also proposes possible causes, consequences, and severity levels for these failure modes.

**Failure Scenario Analyzer:** The design of the Failure Scenario Analyzer involves careful consideration regarding the reward function for the reinforcement learning task of identifying fault injection actions that lead to failure scenarios. Rewarding the agent solely for finding faults that lead to failure scenarios at specific times might lead to over-exploitation of obvious fault injections. To address this, we propose a reward design that scales based on two factors:

1. The less likely a component transitions into the selected fault mode at the time the agent injects its fault, the less reward will be received.
2. The more severe a resulting failure scenario is, the more reward the agent will receive.

By weighting the probability of the fault occurrence with the user-defined failure scenario severity, we derive a reward value that reflects how likely and severe the scenarios can be. As a result, the agent is incentivized to find failure scenarios that are both likely to occur and lead to the most catastrophic failures.


To identify and analyze the critical failure modes of the Franka Emika Panda robotic system with the given specifications, let's consider the following high-level failure modes and their potential severities:

**1. Object Drop**

- **Cause:** Gripper malfunction, insufficient grip strength, or faulty object detection.
- **Consequences:** Damaged objects, potential injury to humans in the vicinity, operational downtime.
- **Severity:** High. Object drops can lead to significant damage and safety hazards, particularly in environments where fragile or hazardous materials are handled.

**2. Collision with Humans**

- **Cause:** Inaccurate perception from the camera, failure in control policy, sensor failure, or unexpected human movement.
- **Consequences:** Injury to humans, potential legal issues, and loss of trust in the robotic system.
- **Severity:** Very High. Human safety is paramount, and collisions can lead to severe injuries and significant repercussions for the deploying organization.

**3. Misidentification of Objects**

- **Cause:** Camera or perception system errors, inaccuracies in the training of the RL agent.
- **Consequences:** Incorrect handling of objects, damage to objects or the environment, and failure to complete tasks.
- **Severity:** Medium to High. Misidentification can disrupt workflow and cause damage but is less severe than direct human harm.

**FIGURE 5: Prompt of our RiskGPT model.**

```
>>> [Step 1]: Deploying RiskGPT Knowledge Engines & DL State Feature Processors...

--- [Phase 1]: Initiating Adversarial Reinforcement Learning Optimization Engine ---

--- [Phase 2]: Initializing Genetic Algorithm Breeding Pipeline for Parameter Tuning ---

>>> System Analysis Complete: Generating Graphs and Comprehensive Performance Report...
```

**Figure 6: Dynamic Exploration Performance Search Curve**

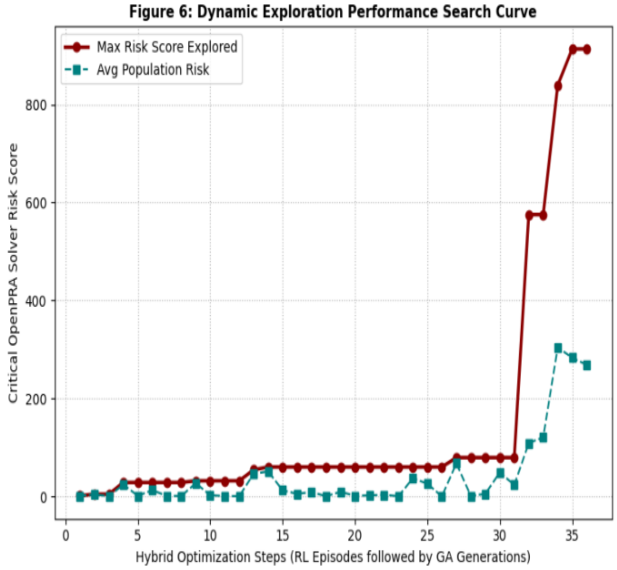



**Figure 7: Spatial End-Effector Trajectory Deviation Profile during 'Move to A'**

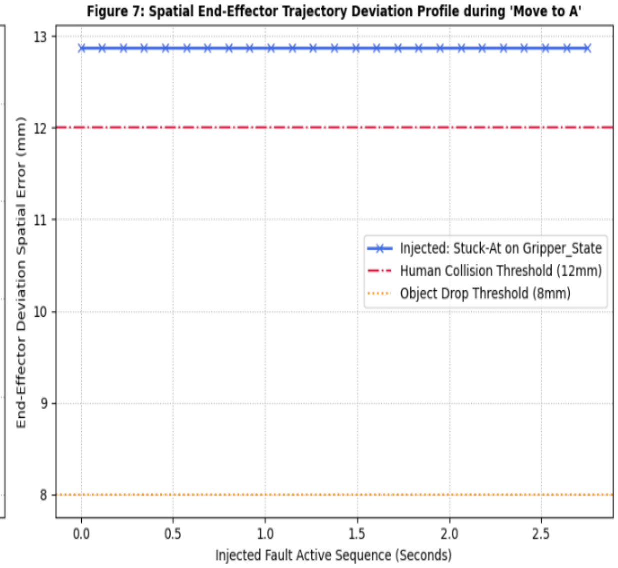



TABLE 2: HYBRID SYSTEM DYNAMIC RISK ASSESSMENT FAILURE SCENARIO MATRIX

| Index | Fault Injected | Target Hardware | Active Skill Phase | Triggered Mode | Risk Score |
|---|---|---|---|---|---|
| 1 | Stuck-At | Gripper_State | Move to A | Collision_with_Human | 913.22 |
| 2 | Stuck-At | Gripper_State | Carry to B | Object_Drop | 781.02 |
| 3 | Stuck-At | Gripper_State | Pick | Object_Drop | 416.25 |
| 4 | Packet_Loss | Gripper_State | Carry to B | Object_Drop | 324.09 |
| 5 | Stuck-At | Camera_Image | Object Detection | Misidentification | 79.27 |
| 6 | Stuck-At | Camera_Image | Place | Misidentification | 74.66 |

## 6. CONCLUSION

In this paper, we introduced the concept of a novel approach that enables the assessment of vulnerabilities in black-box policies, specifically targeting potential failures and comprehensive risk assessment for AI-controlled robotic systems. This method will not only identifies vulnerabilities but also informs the retraining of the original reinforcement learning agent that controls the robotic system, thereby enhancing system safety. For deterministic programmed robotic systems, this approach can provide software developers with insights into specific vulnerabilities. Additionally, we presented a real-world example on which we will demonstrate and evaluate the applicability of our proposed method.

Moreover, we showcased the current status of our research by highlighting the challenges and preliminary results. We discussed the development of our RiskGPT model for identifying critical failure modes and outlined the initial results obtained from applying this model to the Franka Emika Panda robot. Furthermore, we explored the design considerations for the Failure Scenario Analyzer, emphasizing the importance of a well-calibrated reward function in reinforcement learning tasks.

In this paper, we introduce a concept of a new approach, that enables the dynamic risk assessment of dynamic or unknown control policies including AI-controlled robotic systems. The novel approach includes five key methods: Data Logging, Skill

Detection, Behavioral Analysis, Risk Model Generation, and Risk Model Solver.

The approach primarily addresses classical hardware failure. However, we are currently working on an approach that uses large language models to automatically find failure modes and analyze these failure modes with reinforcement learning. The reinforcement learning-based analysis will allow us to understand which events or sequences of events lead to certain failure modes. Events include faults (e.g. bitflip or noise) that will be injected into the system with fault injection methods. The approach will help us to assess the risk of the software parts of the system more precisely. We are working on the realization of this approach in parallel. **"Furthermore, Genetic Algorithms can be integrated with Reinforcement Learning to optimize fault injection parameters and identify critical failure scenarios, thereby enhancing dynamic risk assessment and improving the safety and reliability of AI-controlled robotic systems."**

**Future Scope:** Future work will investigate the integration of Genetic Algorithms with Reinforcement Learning to improve the exploration of fault injection parameters and accelerate the discovery of critical failure scenarios. If given more time, what would you improve? **"If given more time, I would increase the number of real-world experiments, provide more domain-specific knowledge to RiskGPT, and further develop the hybrid RL+GA optimization framework to a production-level system."**


## ACKNOWLEDGMENTS

This work has been partly funded by the German Federal Ministry of Economic Affairs and Climate Action (Bundesministerium für Wirtschaft und Klimaschutz, BMWK) under the project "Software-defined Manufacturing für die Fahrzeug- und Zulieferindustrie" (SDM4FZI, funding code 13IK001ZE) and by the Bundesanstalt für Arbeitsschutz und Arbeitsmedizin (BAuA, Germany) funds, project F 2497.


## REFERENCES


[1] Lin, Long-Ji. *Reinforcement learning for robots using neural networks*. Carnegie Mellon University (1992).

[2] Ibarz, Julian, Tan, Jie, Finn, Chelsea, Kalakrishnan, Mrinal, Pastor, Peter and Levine, Sergey. "How to train your robot with deep reinforcement learning: lessons we have learned." *The International Journal of Robotics Research* Vol. 40 No. 4-5 (2021): pp. 698–721.

[3] Kim, Junyung, Shah, Asad Ullah Amin and Kang, Hyun Gook. "Dynamic risk assessment with bayesian network and clustering analysis." *Reliability Engineering & System Safety* Vol. 201 (2020): p. 106959.

[4] Kalantarnia, Maryam, Khan, Faisal and Hawboldt, Kelly. "Dynamic risk assessment using failure assessment and Bayesian theory." *Journal of Loss Prevention in the Process Industries* Vol. 22 No. 5 (2009): pp. 600–606.

[5] Schneider, Daniel, Reich, Jan, Adler, Rasmus and Liggesmeyer, Peter. "Dynamic Risk Management in Cyber Physical Systems." *arXiv preprint arXiv:2401.13539* (2024).

[6] Kaul, Thorben, Meyer, Tobias and Sextro, Walter. "Integrated model for dynamics and reliability of intelligent mechatronic systems." *European safety and reliability conference (esrel2015). Taylor and Francis*. 2015.

[7] Reich, Jan and Trapp, Mario. "SINADRA: towards a framework for assurable situation-aware dynamic risk assessment of autonomous vehicles." *2020 16th European dependable computing conference (EDCC):* pp. 47–50. 2020. IEEE.

[8] Vicentini, Federico, Askarpour, Mehrnoosh, Rossi, Matteo G and Mandrioli, Dino. "Safety assessment of collaborative robotics through automated formal verification." *IEEE Transactions on Robotics* Vol. 36 No. 1 (2019): pp. 42–61.

[9] Inam, Rafia, Raizer, Klaus, Hata, Alberto, Souza, Ricardo, Forsman, Elena, Cao, Enyu and Wang, Shaolei. "Risk assessment for human-robot collaboration in an automated warehouse scenario." *2018 IEEE 23rd International Conference on Emerging Technologies and Factory Automation (ETFA)*, Vol. 1: pp. 743–751. 2018. IEEE.

[10] Grimmeisen, Philipp, Golwalkar, Rucha, Ma, Yuliang and Morozov, Andrey. "Automated and Continuous Risk Assessment for ROS-Based Software-Defined Robotic Systems." *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE):* pp. 1–7. 2023. IEEE.

[11] Grimmeisen, Philipp, Morozov, Andrey, Fabarisov, Tagir, Wortmann, Andreas and Koo, Chee Hung. "Automated Manufacturing." 2022 *IEEE 27th International Conference on Emerging Technologies and Factory Automation (ETFA)*: pp. 1–4. 2022. IEEE.

[12] Grimmeisen, Philipp, Wortmann, Andreas and Morozov, Andrey. "Case study on automated and continuous reliability assessment of software-defined manufacturing based on digital twins." *Proceedings of the 25th International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings*: pp. 511–518. 2022.

[13] Dong, Yi, Zhao, Xingyu and Huang, Xiaowei. "Dependability analysis of deep reinforcement learning based robotics and autonomous systems through probabilistic model checking." *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS):* pp. 5171–5178. 2022. IEEE.

[14] Brunke, Lukas, Greeff, Melissa, Hall, Adam W, Yuan, Zhaocong, Zhou, Siqi, Panerati, Jacopo and Schoellig, Angela P. "Safe learning in robotics: From learning-based control to safe reinforcement learning." *Annual Review of Control, Robotics, and Autonomous Systems* Vol. 5 (2022): pp. 411–444.

[15] Kwon, Ryeonggu, Kwon, Gihwon, Park, Sohee, Chang, Jiyoung and Jo, Suhee. "Applying Quantitative Model Checking to Analyze Safety in Reinforcement Learning." *IEEE Access* (2024).

[16] Camilli, Matteo, Felderer, Michael, Giusti, Andrea, Matt, Dominik T, Perini, Anna, Russo, Barbara and Susi, Angelo. "Towards risk modeling for collaborative AI." *2021 IEEE/ACM 1st Workshop on AI Engineering-Software Engineering for AI (WAIN):* pp. 51–54. 2021. IEEE.

[17] Villani, Valeria, Pini, Fabio, Leali, Francesco and Secchi, Cristian. "Survey on human-robot collaboration in industrial settings: Safety, intuitive interfaces and applications." *Mechatronics* Vol. 55 (2018): pp. 248–266.

[18] Keskar, Nitish Shirish, McCann, Bryan, Varshney, Lav R, Xiong, Caiming and Socher, Richard. "Ctrl: A conditional transformer language model for controllable generation." *arXiv preprint arXiv:1909.05858* (2019).

[19] Brown, Tom, Mann, Benjamin, Ryder, Nick, Subbiah, Melanie, Kaplan, Jared D, Dhariwal, Prafulla, Neelakantan, Arvind, Shyam, Pranav, Sastry, Girish, Askell, Amanda et al. "Language models are few-shot learners." *Advances in neural information processing systems* Vol. 33 (2020): pp. 1877–1901.

[20] Devlin, Jacob, Chang, Ming-Wei, Lee, Kenton and Toutanova, Kristina. "Bert: Pre-training of deep bidirectional transformers for language understanding." *arXiv preprint arXiv:1810.04805* (2018).

[21] Qi, Yi, Zhao, Xingyu, Khastgir, Siddartha and Huang, Xiaowei. "safety analysis in the era of large language models: a case study of STPA using ChatGPT." *arXiv preprint arXiv:2304.01246* (2023).

[22] Diemert, Simon and Weber, Jens H. "Can Large Language Models Assist in Hazard Analysis?" *International Conference on Computer Safety, Reliability, and Security:* pp. 16–30. 2023.

[23] Favier, Anthony, Messioux, Antonin, Guiochet, Jérémie, Fabre, Jean-Charles and Lesire, Charles. "A hierarchical fault tolerant architecture for an autonomous robot." *2020 50th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops (DSN-W)*: pp. 122–129. 2020. IEEE.

[24] Zhao, Zuoquan, Wang, Jiadong, Cao, Jiawei, Gao, Wenchao and Ren, Qinyuan. "A fault-tolerant architecture for mobile robot localization." *2019 IEEE 15th International Conference on Control and Automation (ICCA)*: pp. 584–589. 2019. IEEE.

[25] Morais, Márcio G, Meneguzzi, Felipe R, Bordini, Rafael H and Amory, Alexandre M. "Distributed fault diagnosis for multiple mobile robots using an agent programming language." *2015 International Conference on Advanced Robotics (ICAR)*: pp. 395–400. 2015. IEEE.

[26] Hsiao, Yu-Shun, Wan, Zishen, Jia, Tianyu, Ghosal, Radhika, Mahmoud, Abdulrahman, Raychowdhury, Arijit, Brooks, David, Wei, Gu-Yeon and Reddi, Vijay Janapa. "Mavfi: An end-to-end fault analysis framework with anomaly detection and recovery for micro aerial vehicles." *2023 Design, Automation & Test in Europe Conference & Exhibition (DATE)*: pp. 1–6. 2023. IEEE.

[27] Fabarisov, Tagir, Morozov, Andrey, Mamaev, Ilshat and Grimmeisen, Philipp. "Fidget: Deep learning-based fault injection framework for safety analysis and intelligent generation of labeled training data." *2022 IEEE 27th International Conference on Emerging Technologies and Factory Automation (ETFA)*: pp. 1–6. 2022. IEEE.

[28] Alemzadeh, Homa, Chen, Daniel, Li, Xiao, Kesavadas, Thenkurussi, Kalbarczyk, Zbigniew T and Iyer, Ravishankar K. "Targeted attacks on teleoperated surgical robots: Dynamic model-based detection and mitigation." *2016 46th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*: pp. 395–406. 2016. IEEE.

[29] Li, Xiao, Alemzadeh, Homa, Chen, Daniel, Kalbarczyk, Zbigniew, Iyer, Ravishankar K and Kesavadas, Thenkurussi. "A hardware-in-the-loop simulator for safety training in robotic surgery." *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*: pp. 5291–5296. 2016. IEEE.

[30] Ma, Yuliang, Grimmeisen, Philipp and Morozov, Andrey. "Case Study: ROS-Based Fault Injection for Risk Analysis of Robotic Manipulator." *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)*: pp. 1–6. 2023. IEEE.

[31] Gao, Yunfan, Xiong, Yun, Gao, Xinyu, Jia, Kangxiang, Pan, Jinliu, Bi, Yuxi, Dai, Yi, Sun, Jiawei and Wang, Haofen. "Retrieval-augmented generation for large language models: A survey." *arXiv preprint arXiv:2312.10997* (2023).

[32] Lewis, Patrick, Perez, Ethan, Piktus, Aleksandra, Petroni, Fabio, Karpukhin, Vladimir, Goyal, Naman, Küttler, Heinrich, Lewis, Mike, Yih, Wen-tau, Rocktäschel, Tim et al. "Retrieval-augmented generation for knowledge-intensive nlp tasks." *Advances in Neural Information Processing Systems* Vol. 33 (2020): pp. 9459–9474.

[33] Topsakal, Oguzhan and Akinci, Tahir Cetin. "Creating large language model applications utilizing langchain: A primer on developing llm apps fast." *International Conference on Applied Engineering and Natural Sciences*, Vol. 1. 1: pp. 1050–1056. 2023.

[34] Makoviychuk, Viktor, Wawrzyniak, Lukasz, Guo, Yunrong, Lu, Michelle, Storey, Kier, Macklin, Miles, Hoeller, David, Rudin, Nikita, Allshire, Arthur, Handa, Ankur et al. "Isaac gym: High performance gpu-based physics simulation for robot learning." *arXiv preprint arXiv:2108.10470* (2021).

[35] Open Robotics. "Gazebo Simulator." https://gazebosim.org/home (accessed 2024).

[36] Todorov, Emanuel, Erez, Tom and Tassa, Yuval. "MuJoCo: A physics engine for model-based control." *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*: pp. 5026–5033. 2012. IEEE. DOI 10.1109/IROS.2012.6386109.

[37] Moerland, Thomas M, Broekens, Joost, Plaat, Aske, Jonker, Catholijn M et al. "Model-based reinforcement learning: A survey." *Foundations and Trends® in Machine Learning* Vol. 16 No. 1 (2023): pp. 1–118.

[38] Schulman, John, Wolski, Filip, Dhariwal, Prafulla, Radford, Alec and Klimov, Oleg. "Proximal policy optimization algorithms." *arXiv preprint arXiv:1707.06347* (2017).

[39] Mnih, Volodymyr, Kavukcuoglu, Koray, Silver, David, Graves, Alex, Antonoglou, Ioannis, Wierstra, Daan and Riedmiller, Martin. "Playing atari with deep reinforcement learning." *arXiv preprint arXiv:1312.5602* (2013).

[40] Burda, Yuri, Edwards, Harrison, Storkey, Amos and Klimov, Oleg. "Exploration by random network distillation." *arXiv preprint arXiv:1810.12894* (2018).

[41] Kamthe, Sanket and Deisenroth, Marc. "Data-efficient reinforcement learning with probabilistic model predictive control." *International conference on artificial intelligence and statistics*: pp. 1701–1710. 2018. PMLR.

[42] Grimmeisen, Philipp, Karimov, Artur, Diaconeasa, Mihai A and Morozov, Andrey. "Demonstration of a limited scope probabilistic risk assessment for autonomous warehouse robots with openpra." *ASME International Mechanical Engineering Congress and Exposition*, Vol. 85697: p. V013T14A030. 2021. American Society of Mechanical Engineers.

[43] Höfer, Sebastian, Bekris, Kostas, Handa, Ankur, Gamboa, Juan Camilo, Mozifian, Melissa, Golemo, Florian, Atkeson, Chris, Fox, Dieter, Goldberg, Ken, Leonard, John et al. "Sim2Real in robotics and automation: Applications and challenges." *IEEE transactions on automation science and engineering* Vol. 18 No. 2 (2021): pp. 398–400.

[44] Julian Ibarz et al. "How to train your robot with deep reinforcement learning: lessons we have learned". In: *The International Journal of Robotics Research* 40.4-5 (2021), pp. 698–721.

[45] Enno Ruijters and Mariëlle Stoelinga. "Fault tree analysis: A survey of the state-of-the-art in modeling, analysis and tools". In: *Computer science review* 15 (2015), pp. 29–62.

[46] Man Cheol Kim. "Reliability block diagram with general gates and its application to system reliability analysis". In: *Annals of Nuclear Energy* 38.11 (2011), pp. 2456–2461.

[47] Andrea Bobbio et al. "Improving the analysis of dependable systems by mapping fault trees into Bayesian networks". In: *Reliability Engineering & System Safety* 71.3 (2001), pp. 249–260.

[48] Norman B Fuqua. "The applicability of markov analysis methods to reliability, maintainability, and safety". In: *Selected Topic in Assurance Related Technologies (START)* 2.10 (2003), pp. 1–8.

[49] Falko Bause and Pieter S Kritzinger. *Stochastic petri nets*. Vol. 1. Vieweg Wiesbaden, 2002.

[50] Andrey Morozov and Klaus Janschek. "Dual graph error propagation model for mechatronic system analysis". In: *IFAC Proceedings Volumes* 44.1 (2011), pp. 9893–9898.

[51] Marta Kwiatkowska, Gethin Norman, and David Parker. "PRISM: Probabilistic symbolic model checker". In: *International Conference on Modelling Techniques and Tools for Computer Performance Evaluation*. Springer. 2002, pp. 200–204.

[52] Philipp Grimmeisen et al. "Demonstration of a Limited Scope Probabilistic Risk Assessment for Autonomous Warehouse Robots With OpenPRA". In: *ASME International Mechanical Engineering Congress and Exposition*. Vol. 85697. American Society of Mechanical Engineers. 2021, V013T14A030.